Sampling Methods: A Primer

Training handout — held back for live generation.

1. Probability Sampling

A probability sample is one in which every unit in the population has a known, non-zero chance of selection. That known probability is what permits design-based inference: the estimate carries a measurable sampling error, and a confidence interval means what it claims to mean. A sample chosen for convenience carries no such guarantee, however large it is.

The sampling frame is the list from which units are drawn. Frame quality bounds everything downstream. A frame that omits recently constructed urban blocks will under-represent new migrants no matter how carefully the sample is selected from it, and no amount of weighting recovers units that were never eligible for selection.

2. Stratification

Stratification divides the population into mutually exclusive and exhaustive groups, called strata, and draws an independent sample within each. Strata are formed on variables known for the whole frame, such as state, sector and district. Because sampling is independent within each stratum, the design guarantees representation of every stratum, which simple random sampling from the whole population does not.

Stratification reduces variance when strata are internally homogeneous with respect to the variable being measured and differ from one another. Allocation across strata may be proportional to stratum size, or it may be optimal allocation, which additionally accounts for the variability within each stratum and the cost of enumerating there.

3. Multistage Designs and Weights

Large household surveys use multistage designs. First-stage units, typically villages or urban blocks, are selected with probability proportional to size. Households are then selected within each chosen first-stage unit. This concentrates field work geographically and makes national coverage affordable.

The design weight of a unit is the reciprocal of its overall selection probability, and it is what makes a sample estimate represent the population. Weights are then adjusted for non-response and calibrated against known population totals from the census or from projections. The design effect measures how much the variance of an estimate has been inflated relative to simple random sampling of the same size, and it is the number to quote when someone asks why the sample is larger than a textbook formula suggests.